Cross-Domain Precision Data Inventory

Raw Database

Registry: [HOWL-DATA-1-2026]

Series Path: [HOWL-DATA-1-2026] → [HOWL-DATA-2-2026]

DOI: 10.5281/zenodo.19665835

Date: April 1 2026

Domain: Cross Domain Data

Status: Documentation

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Abstract

This paper presents a curated reference database of precision-measured constants, defined standards, and analytical results from every domain where $R_2 = \pi/4$ appears in the governing equation. The inventory spans 17 domain categories and 250+ individual entries, each carrying the original measured or defined value, its source, and the R_2/R_4 equation it participates in. The database serves as the data foundation for the HOWL series: every claim, test, and computation in the series draws from values catalogued here. DATA-2 (forthcoming) will transform applicable entries into big-integer rational representations and test them against the Q335 basis.

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1. Purpose and Scope

The HOWL series established two structural results. First, $R_2 = \pi/4$ appears in the modulus of every circular-to-rectilinear conversion across nine engineering domains and nine physics domains (MATH-1, PHYS-11). Second, the framework determines geometric structure (Level 1) but not continuous parameter values (Level 2), with 72/72 PSLQ tests confirming algebraic independence of measured constants from the transcendental basis (DISC-6 through DISC-9).

Both results require data. The structural claim requires the governing equations. The independence claim requires the measured values at full available precision. This paper collects both: the equations and the numbers.

The scope is breadth-first. Every domain where R_2 or R_4 enters a precision equation is included, with the best available source data. Entries are classified by data type:

- **Type E** (Exact): defined by international agreement, zero uncertainty

- **Type S** (Standard nominal): defined by industry specification, exact to published digits
- **Type A** (Analytical): derived from theory, exact in closed form
- **Type M** (Measured): experimentally determined, uncertainty quoted

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2. SI Fundamental Constants (Type E)

These seven values anchor the entire measurement system. Since the 2019 SI redefinition, all have zero uncertainty. Their rational representations are exact.

ID	Constant	Symbol	Value	Unit	Fraction
A1	Speed of light	c	299792458	m/s	299792458/1
A2	Planck constant	h	$6.62607015 \times 10^{-34}$	J·s	$662607015/10^{42}$
A3	Elementary charge	e	$1.602176634 \times 10^{-19}$	C	$1602176634/10^{28}$
A4	Boltzmann constant	k B	1.380649×10^{-23}	J/K	$1380649/10^{29}$
A5	Avogadro constant	N A	$6.02214076 \times 10^{23}$	mol^{-1}	602214076×10^{15}
A6	Cs-133 hyperfine frequency	$\Delta\ \nu$ Cs	9192631770	Hz	9192631770/1
A7	Luminous efficacy	K cd	683	lm/W	683/1

R_2 connection: $h = 2\,\pi\,\hbar = 8R_2\hbar$. Every appearance of $\hbar$ in physics carries an implicit factor of $1/(8R_2)$ relative to h. The SI chose to define h (not $\hbar$), which embeds the geometric content into the defined constant.

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3. Derived Exact Constants (Type E, from SI)

These are computed exactly from the seven SI constants. Where π enters, the value is exact but irrational.

ID	Constant	Symbol	Expression	Numerical Value	R_2/R_4 Form
B1	Reduced Planck	$\hbar$	$h/(2\pi)$	$1.054571817\ldots \times 10^{-34}$ J·s	$h/(8R_2)$
B2	Josephson constant	K J	2e/h	$483597.848\ldots \times 10^{9}$ Hz/V	$2e/(8R_2\hbar)$
B3	von Klitzing constant	R K	h/e^2	$25812.80745\ldots$ Ω	$8R_2\hbar/e^2$
B4	Flux quantum	Φ_0	h/(2e)	$2.067833848\ldots \times 10^{-15}$ Wb	$4R_2\hbar/e$
B5	Conductance quantum	G_0	$2e^2/h$	$7.748091729\ldots \times 10^{-5}$ S	$e^2/(4R_2\hbar)$
B6	Stefan-Boltzmann	σ	$2\pi^5k^4/(15h^3c^2)$	$5.670374419\ldots \times 10^{-8}$ $W/(m^2K^4)$	$32R_4\cdot\pi^3k^4/(15h^3c^2)$
B7	First radiation const	c_1	$2\pi hc^2$	$3.741771852\ldots \times 10^{-16}$ $W\cdot m^2$	$8R_2hc^2$
B8	Second radiation const	c_2	hc/k B	$1.438776877\ldots \times 10^{-2}$ m·K	exact rational

R_2 cancellation identities:

- K_J × R_K = $2e/h \times h/e^2 = 2/e$. R_2 cancels exactly.
- G_0 × R_K = $2e^2/h \times h/e^2 = 2$. R_2 cancels exactly.
- $R\infty = \alpha^2 m\_ec/(2h) = \alpha^2 m\_ec/(16R_2\hbar)$. R_2 cancels because $2h = 16R_2\hbar$ and $h = 8R_2\hbar$ fold into the ratio.

The most precisely measured constants in nature are those where R_2 drops out. The Rydberg constant at 13 digits and the electron g-factor at 13 digits are both R_2-independent.

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4. Measured Fundamental Constants (Type M, CODATA 2022)

These are the measured inputs to the Standard Model. Each carries an experimentally determined uncertainty. The digits shown are the full CODATA 2022 precision.

ID	Constant	Symbol	Value	Rel. Uncertainty	Digits
C1	Fine structure constant	α^{-1}	137.035999177(21)	1.5×10^{-10}	12
C2	Electron mass	m_e	0.51099895069(16) MeV	3.1×10^{-10}	11
C3	Muon mass	m_μ	105.6583755(23) MeV	2.2×10^{-8}	10
C4	Tau mass	m_τ	1776.86(12) MeV	6.8×10^{-5}	6
C5	Proton mass	m_p	938.27208943(29) MeV	3.1×10^{-10}	11
C6	Proton-electron mass ratio	m_p/m_e	1836.15267343(32)	1.7×10^{-10}	13
C7	Rydberg constant	R_∞	10973731.568157(12) m^{-1}	1.1×10^{-12}	13
C8	Bohr radius	a_0	5.29177210544(82) $\times 10^{-11}$ m	1.6×10^{-10}	12
C9	Electron g-factor anomaly	a_e	0.00115965218059(13)	1.1×10^{-10}	12
C10	Muon g-factor anomaly	a_μ	0.00116592059(22)	1.9×10^{-7}	8
C11	Weak mixing angle	$\sin^2\theta_W$	0.23122(4)	1.7×10^{-4}	5
C12	Strong coupling	$\alpha_s(M_Z)$	0.1180(9)	7.6×10^{-3}	4

R_2 connection: α enters every QED calculation. The anomalous magnetic moment a_e = $A_1(\alpha/\pi) + A_2(\alpha/\pi)^2 + \ldots$ where $A_1 = 1/2$, and every power of $\alpha/\pi = \alpha/(4R_2)$ carries R_2. The QED perturbation series is an expansion in $\alpha/(4R_2)$.

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5. Electroweak Observables (Type M, LEP/PDG)

These are the precision measurements from LEP, SLD, and the Tevatron/LHC that constrain the electroweak sector.

ID	Observable	Value	Precision	Source
C13	Z boson mass	M Z = 91187.6(2.1) MeV	23 ppm	LEP EWWG
C14	Z boson width	Γ Z = 2495.2(2.3) MeV	0.09%	LEP EWWG
C15	W boson mass	M W = 80369.2(13.3) MeV	0.017%	PDG 2024 avg
C16	Top quark mass	m t = 172.57(29) GeV	0.17%	CMS+ATLAS
C17	Higgs boson mass	m H = 125.20(11) GeV	0.09%	ATLAS+CMS
C18	Hadronic cross section	σ^0 had = 41.481(33) nb	0.08%	LEP
C19	R l ratio	R l = 20.767(25)	0.12%	LEP
C20	R b ratio	R b = 0.21629(66)	0.31%	LEP
C21	Forward-backward asymmetry	A FB^l = 0.0171(10)	5.8%	LEP
C22	Left-right asymmetry	A l(SLD) = 0.1513(21)	1.4%	SLD
C23	Number of neutrinos	N ν = 2.9840(82)	0.27%	LEP (consistent with 3)
C24	Fermi constant	G F = 1.1663788(6) $\times 10^{-5}$ GeV^{-2}	5.1×10^{-7}	Muon lifetime

R_2 connection: every partial width Γ_f = (G_F M_Z^3)/(6 π √2) × (v_f^2 + a_f^2) × N_c × corrections contains 1/ π = 1/(4R_2) in the denominator. The electroweak mixing enters through v_f = T_3 − 2Q_f sin^{2}θ_W, where T_3 and Q_f are integers or simple fractions.

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6. Quark Masses and CKM Parameters (Type M, PDG 2024/Lattice)

ID	Parameter	Value	Source	Notes
C25	Up quark mass	m_u = 2.16(7) MeV	PDG 2024	MS-bar at 2 GeV
C26	Down quark mass	m_d = 4.70(7) MeV	PDG 2024	MS-bar at 2 GeV
C27	Strange quark mass	m_s = 93.5(8) MeV	PDG 2024	MS-bar at 2 GeV
C28	Charm quark mass	m_c = 1.273(4) GeV	PDG 2024	MS-bar at m_c
C29	Bottom quark mass	m_b = 4.183(7) GeV	PDG 2024	MS-bar at m_b
C30	Top quark mass (pole)	m_t = 172.57(29) GeV	CMS+ATLAS	Direct measurement
C31	CKM $\sin\theta_{12}$	0.22501(67)	PDG 2024	Cabibbo angle
C32	CKM $\sin\theta_{23}$	0.04182(85)	PDG 2024	
C33	CKM $\sin\theta_{13}$	0.003685(93)	PDG 2024	
C34	Lattice m_c/m_s	11.783(25)	FLAG 2023	5 sig figs
C35	Lattice m_b/m_c	4.578(8)	FLAG 2023	4 sig figs
C36	Lattice m_u/m_d	0.485(19)	FLAG 2023	3 sig figs

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7. Nuclear and Hadron Masses (Type M, CODATA/PDG)

ID	Particle	Mass (MeV)	Uncertainty	Digits
C37	Proton	938.27208943(29)	3.1×10^{-10}	11
C38	Neutron	939.56542194(48)	5.1×10^{-10}	11
C39	n − p mass difference	1.29333251(38)	2.9×10^{-7}	7
C40	Pion (charged)	139.57039(18)	1.3×10^{-6}	9
C41	Pion (neutral)	134.9770(5)	3.7×10^{-6}	7
C42	Kaon (charged)	493.677(16)	3.2×10^{-5}	6
C43	Deuteron	1875.61294500(58)	3.1×10^{-10}	11
C44	He-4	3727.3794118(12)	3.2×10^{-10}	11
C45	Deuteron binding energy	2.22456614(42)	1.9×10^{-7}	7

8. Atomic Spectroscopy and Clock Frequencies (Type M)

ID	Transition	Frequency (Hz)	Precision	Source
C46	H 1S-2S	2466061413187018(11)	4.5×10^{-15}	Parthey et al. 2011
C47	H hyperfine 1S	1420405751768(1) mHz	7.0×10^{-13}	H maser
C48	^{87}Sr clock	429228004229873.0(2)	4.7×10^{-16}	BIPM CIPM
C49	^{171}Yb clock	518295836590863.6(3)	5.8×10^{-16}	BIPM CIPM
C50	$^{27}Al^{+}$ clock	~1121015393207857.3	$\sim 10^{-18}$	NIST
C51	Lamb shift $2S\text{-}2P_1/_2$	1057845.0(9) kHz	8.5×10^{-7}	Spectroscopy
C52	Proton charge radius	0.84075(64) fm	7.6×10^{-4}	Muonic hydrogen

R_2 connection: the hydrogen 1S-2S frequency depends on R∞, which is R_2-independent (Section 3). The Lamb shift depends on $\alpha(Z\alpha)^4 \ln(\alpha)$, where every $\alpha/\pi = \alpha/(4R_2)$. Clock frequencies are integers by definition of the second via $\Delta\nu$ _Cs.

9. Optical Storage (Type S)

All values are standard nominals from the specification documents. The R_2 connection: every disc area is $R_2 \times d^2$, and the diffraction-limited spot size is $1.22\lambda/NA = (3.8317/\pi)\lambda/NA$, where 3.8317 is the first zero of J_1.

9.1 CD (Red Book / ECMA-130)

ID	Parameter	Value	Fraction	R_2 Equation
D1	Laser wavelength	780 nm	39/50000000 m	Spot = $1.22\lambda/NA$
D2	Track pitch	1.60 μm	1/625000 m	—
D3	Pit width	500 nm	1/2000000 m	—
D4	Pit depth	125 nm	1/8000000 m	$= \lambda/(4n)$, quarter-wave
D5	Min pit length	830 nm	83/100000000 m	—
D6	Disc diameter	120 mm	3/25 m	Area = $R_2 \times (0.12)^2$
D7	Disc thickness	1.2 mm	3/2500 m	—

ID	Parameter	Value	Fraction	R_2 Equation
D8	Sampling rate	44100 Hz	44100/1	$= 2^2 \times 3^2 \times 5^2 \times 7^2$
D9	Sector size (raw)	2352 bytes	2352/1	$= 2^4 \times 3 \times 7^2$
D10	Sector size (data)	2048 bytes	2048/1	$= 2^{11}$
D11	Scanning velocity	1.2–1.4 m/s	6/5 to 7/5	—
D12	Numerical aperture	~0.45	9/20	Spot = $1.22 \times 780/0.45 \approx 2.1$ μm
D13	Capacity	700 MB	$700 \times 10^6/1$	—

9.2 DVD (ECMA-267)

ID	Parameter	Value	Fraction	R_2 Equation
D14	Laser wavelength	650 nm	13/20000000 m	Spot = $1.22\lambda/NA$
D15	Track pitch	0.74 μm	37/50000000 m	—
D16	Pit width	320 nm	1/3125000 m	—
D17	Pit depth	120 nm	3/25000000 m	—
D18	Min pit length	400 nm	1/2500000 m	—
D19	Channel bit length	133.3 nm	1333/10000000000 m	—
D20	Disc substrate	0.6 mm	3/5000 m	—
D21	Numerical aperture	0.60	3/5	Spot = $1.22 \times 650/0.60 \approx 1.3$ μm
D22	Sector size	2048 bytes	2^{11}	—
D23	Capacity (SL)	4.7 GB	$47/10 \times 10^9$	—

9.3 Blu-ray (BD White Paper)

ID	Parameter	Value	Fraction	R_2 Equation
D24	Laser wavelength	405 nm	81/200000000 m	Spot = $1.22\lambda/NA$
D25	Track pitch	0.320 μm	1/3125000 m	—
D26	Min pit length (2T)	149 nm	149/1000000000 m	—
D27	Cover layer	0.100 mm	1/10000 m	—

ID	Parameter	Value	Fraction	R_2 Equation
D28	Numerical aperture	0.85	17/20	Spot = 1.22 × 405/0.85 ≈ 582 nm
D29	Spot size	580 nm	29/50000000 m	= 1.22λ/NA
D30	Capacity (SL)	25 GB	25×10^9	—
D31	Disc tilt spec	0.35°	7/2000 rad (approx)	—

Disc area summary (all $R_2 \times d^2$):

- All three formats: diameter = 120 mm, area = $R_2 \times (120)^2 = R_2 \times 14400$ mm^2 = 11309.73 mm^2

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10. RAM and Memory (Type S, JEDEC)

All timing specifications are exact rationals derived from crystal oscillator frequencies. CAS latencies are exact integers.

10.1 Transfer Rates and Clock Periods

ID	Standard	Transfer Rate	Clock Period (t CK)	Unit Interval	Fraction (UI)
E1	DDR4-1600	1600 MT/s	1/800 MHz = 1.250 ns	1/1600 MHz = 0.625 ns	1/1600000000 s
E2	DDR4-2400	2400 MT/s	1/1200 MHz = 0.833 ns	1/2400 MHz = 0.417 ns	1/2400000000 s
E3	DDR4-3200	3200 MT/s	1/1600 MHz = 0.625 ns	1/3200 MHz = 0.3125 ns	1/3200000000 s
E4	DDR5-4800	4800 MT/s	1/2400 MHz = 0.417 ns	1/4800 MHz = 0.208 ns	1/4800000000 s
E5	DDR5-5600	5600 MT/s	1/2800 MHz	1/5600 MHz	1/5600000000 s
E6	DDR5-6400	6400 MT/s	1/3200 MHz = 0.3125 ns	1/6400 MHz = 0.15625 ns	1/6400000000 s

10.2 Bus Architecture

ID	Parameter	Value	Type
E7	Data bus width	64 bits	Exact integer
E8	ECC bus width	72 bits	Exact integer
E9	Burst lengths	2, 4, 8, 16	Exact integers (powers of 2)
E10	DDR4 voltage	1.2 V	Exact rational = 6/5
E11	DDR5 voltage	1.1 V	Exact rational = 11/10
E12	DRAM cell area	$6F^2$ (standard), $4F^2$ (emerging)	Exact integer $\times$ F^2
E13	SRAM cell area	~120 F^2	Approximate

R_2 connection: the crystal oscillator that drives every memory system has frequency f = $1/(2\ \pi\ \sqrt{(LC)}) = 1/(8R_2\sqrt{(LC)})$. The $8R_2$ is buried in the oscillator formula and inherited by every timing specification.

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11. SSD and Storage Interfaces (Type S)

ID	Interface	Line Rate	Unit Interval	Fraction (UI)
F1	SATA I	1.5 Gbit/s	666.67 ps	1/1500000000 s
F2	SATA II	3.0 Gbit/s	333.33 ps	1/3000000000 s
F3	SATA III	6.0 Gbit/s	166.67 ps	1/6000000000 s
F4	PCIe Gen 3	8.0 GT/s	125.00 ps	1/8000000000 s
F5	PCIe Gen 4	16.0 GT/s	62.50 ps	1/16000000000 s
F6	PCIe Gen 5	32.0 GT/s	31.25 ps	1/32000000000 s
F7	NVMe LBA (legacy)	512 bytes	—	2^9
F8	NVMe LBA (AF)	4096 bytes	—	2^{12}
F9	NAND page size (typ)	16384 bytes	—	2^{14}
F10	3D NAND layers (2024)	200+	—	Integer
F11	MRAM half-pitch (record)	20.5 nm	—	IEDM 2024

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12. Wire, Cable, and Conductors (Type E/S)

12.1 Unit Definitions (Type E)

ID	Definition	Value	Exact Fraction
G1	1 inch	25.4 mm	127/5 mm
G2	1 mil	0.0254 mm	127/5000 mm
G3	1 circular mil	area of circle with d = 1 mil	$R_2 \times (127/5000000)^2$ m^2

12.2 AWG System (Type A/S)

ID	Property	Value	Notes
G4	AWG formula	d n = 0.005” × 92^((36−n)/39)	Irrational (92^(1/39))
G5	Diameter ratio per gauge	92^(1/39) ≈ 1.12293	~10.6% per step
G6	Area ratio per 3 gauges	2.000 (exact, by construction)	3 gauges = double area
G7	AWG 0000 (4/0) diameter	0.4600” = 11.684 mm	4 sig figs per ASTM
G8	AWG 36 diameter	0.0050” = 0.127 mm	Exact anchor point
G9	AWG 10 diameter	0.1019” = 2.588 mm	4 sig figs
G10	AWG 12 diameter	0.0808” = 2.053 mm	Area = $R_2 \times (2.053)^2 = 3.31$ mm^2
G11	AWG 14 diameter	0.0641” = 1.628 mm	Area = $R_2 \times (1.628)^2 = 2.08$ mm^2
G12	AWG tabulation precision	4 significant figures	ASTM B258-02

R_2 equation for every AWG entry: Area_actual = $R_2 \times d^2 = (\pi/4) \times d^2$. The circular mil system defines Area_cmil = d^2(mils) to AVOID computing R_2. The conversion is Area_actual = $R_2 \times$ Area_cmil $\times$ (mil)^2. Every wire sizing calculation since 1857 uses R_2.

12.3 IEC Conductor Sizes (Type S)

ID	Nominal CSA (mm^2)	Fraction
G13	0.50	1/2
G14	0.75	3/4
G15	1.00	1/1
G16	1.50	3/2
G17	2.50	5/2

ID	Nominal CSA (mm^2)	Fraction
G18	4.00	4/1
G19	6.00	6/1
G20	10.0	10/1

Source: IEC 60228.

12.4 Conductor Properties (Type E/M)

ID	Property	Value	Type	Source
G21	Copper conductivity (100% IACS)	5.8001×10^7 S/m	Defined reference	IACS
G22	Standard impedances	50 Ω, 75 Ω, 110 Ω	Exact integers	RF/telecom
G23	Copper resistivity at 20°C	1.7241×10^{-8} Ω·m	Measured, 4 sf	Handbook

R_2 in wire resistance: $R = \rho L/A = \rho L/(R_2 d^2)$. Every wire resistance calculation divides by R_2.

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13. Acoustics and Audio (Type E/S/M)

13.1 Reference Values (Type E)

ID	Reference	Value	Fraction	Source
H1	SPL reference pressure	20 μPa	1/50000 Pa	IEC
H2	Sound power reference	1 pW	$1/10^{12}$ W	IEC
H3	Standard atmosphere	101325 Pa	101325/1	ISO 2533
H4	Standard pitch A4	440 Hz	440/1	ISO 16

13.2 Digital Audio (Type S)

ID	Parameter	Value	Prime Factorization
H5	CD sample rate	44100 Hz	$2^2 \times 3^2 \times 5^2 \times 7^2$
H6	Studio rate	48000 Hz	$2^7 \times 3 \times 5^3$
H7	High-res rate	96000 Hz	$2^8 \times 3 \times 5^3$
H8	Ultra high-res rate	192000 Hz	$2^9 \times 3 \times 5^3$

ID	Parameter	Value	Prime Factorization
H9	CD bit depth	16 bits	2^4
H10	Studio bit depth	24 bits	$2^3 \times 3$
H11	Nominal impedances	4, 8, 16 Ω	Powers of 2

13.3 Just Intonation (Type A — exact rationals)

ID	Interval	Ratio
H12	Octave	2/1
H13	Perfect fifth	3/2
H14	Perfect fourth	4/3
H15	Major third	5/4
H16	Minor third	6/5
H17	Major sixth	5/3
H18	Minor seventh	9/5

13.4 Equal Temperament (Type A — irrational)

ID	Property	Value	Notes
H19	Semitone ratio	2^(1/12) ≈ 1.059463	Irrational
H20	Pythagorean comma	$(3/2)^{12} / 2^7$ = 531441/524288	Exact rational

13.5 Speaker Parameters (Type M)

ID	Parameter	R_2 Equation	Typical Range	Precision
H21	Cone area Sd	Sd = $R_2 \times$ d eff^2	50–2000 cm^2	± 5%
H22	Resonant frequency Fs	Via Helmholtz: f = $(c/8R_2)\sqrt{(A/lV)}$	20–200 Hz	± 2 Hz
H23	Total Q factor Qts	Dimensionless	0.2–1.2	± 10%
H24	Equivalent volume Vas	Involves air compliance	5–500 liters	± 10%
H25	Voice coil resistance Re	Direct measurement	2–8 Ω	± 5%

R_2 connection: every circular speaker cone has area Sd = $R_2 \times d^2$. The Helmholtz resonance of a ported enclosure is f = $(c/(8R_2))\sqrt{(S\_port/(l\_eff \times V\_box))}$, containing $8R_2 = 2\pi$ in the denominator. Sound intensity from a point source falls as $I = P/(4\pi r^2) = P/(16R_2r^2)$.

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14. RF, Telecom, and Signal Processing (Type E/S/A)

14.1 Defined Standards

ID	Parameter	Value	Fraction	Source
I1	T1 carrier rate	1.544 Mbps	193/125 Mbit/s	Telecom
I2	Ethernet min frame	64 bytes	2^6	IEEE 802.3
I3	Ethernet max frame	1518 bytes	1518/1	IEEE 802.3
I4	Standard baud: 9600	9600 Bd	$2^7 \times 3 \times 5^2$	RS-232
I5	Standard baud: 115200	115200 Bd	$2^8 \times 3^2 \times 5^2$	RS-232
I6	5G NR subcarrier spacing	15×2^n kHz (n=0..4)	15, 30, 60, 120, 240 kHz	3GPP
I7	GPS L1 frequency	1575.42 MHz	154×10.23 MHz	GPS ICD
I8	GPS L2 frequency	1227.60 MHz	120×10.23 MHz	GPS ICD
I9	GPS base frequency	10.23 MHz	1023/100 MHz	GPS ICD
I10	GPS seconds/week	604800	$2^5 \times 3^3 \times 5^2 \times 7$	GPS

14.2 R_2 Equations in RF

ID	Equation	R_2 Form	Industrial Use
I11	Antenna effective aperture	$A\ eff = G\lambda^2/(4\pi) = G\lambda^2/(16R_2)$	Every antenna calibration
I12	Free-space path loss	$FSPL = (4\pi d/\lambda)^2 = (16R_2 d/\lambda)^2$	Every link budget
I13	$\pi/4$-DQPSK modulation	Rotates constellation by $\pi/4 = R_2$	IS-136, TETRA

ID	Equation	R_2 Form	Industrial Use
I14	Friis transmission	$P r/P t = G tG r(\lambda/(4 \pi d))^2 = G tG r(\lambda/(16R_2d))^2$	Radar, comm
I15	Radar cross section	$\sigma = 4 \pi A^2/\lambda^2 = 16R_2A^2/\lambda^2$ (flat plate)	Every RCS calculation

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15. Semiconductor and Lithography (Type S/M)

ID	Parameter	Value	R_2 Connection	Source
J1	Standard wafer diameter	300 mm	Area = $R_2 \times (300)^2 = 70685.83$ mm^2	SEMI
J2	Si lattice constant	5.431020511(89) Å	—	CODATA 2022
J3	EUV wavelength	13.5 nm	Spot size via Airy: $1.22\lambda/NA$	ASML
J4	ArF immersion λ	193 nm	Same Airy formula	Industry
J5	DRAM cell area	$6F^2$ (planar), $4F^2$ (VCT)	F is minimum feature size	JEDEC/industry
J6	SRAM cell area	~120 F^2	—	Industry
J7	Rayleigh criterion	$\sin \theta = 1.22\lambda/D$	$1.22 = 3.8317/\pi = j_{11}/\pi$	Optics
J8	Gaussian implant	$N(x) \propto 1/\surd(2 \pi \sigma^2) \exp(-x^2/2\sigma^2)$	$1/\surd(8R_2)$ normalization	Ion implant
J9	Fick's diffusion	erfc contains $1/\surd(\pi t)$	$1/\surd(4R_2t) = 1/(2\surd(R_2t))$	Diffusion

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16. Optics and Photonics (Type A/M)

16.1 Gaussian Beam Equations (all contain R_2)

ID	Formula	Standard Form	R_2 Form
K1	Rayleigh range	$z R = \pi w_0^2/\lambda$	$4R_2w_0^2/\lambda$
K2	Beam divergence (half-angle)	$\theta = \lambda/(\pi w_0)$	$\lambda/(4R_2w_0)$
K3	Beam area at waist	$A = \pi w_0^2$	$4R_2w_0^2$
K4	Beam parameter product	$BPP = \lambda/ \pi$	$\lambda/(4R_2)$
K5	Confocal parameter	$b = 2z R = 2 \pi w_0^2/\lambda$	$8R_2w_0^2/\lambda$
K6	Gouy phase	$\psi(z) = \arctan(z/z R)$	Phase accumulated through R_2

16.2 Fiber Optic Data (Type S/M)

ID	Parameter	SMF-28 Value	R_2 Equation	Source
K7	Mode field diameter (1310 nm)	9.2 ± 0.4 μm	A eff $= R_2 \times MFD^2$	Corning
K8	Mode field diameter (1550 nm)	~10.4 μm	A eff $= R_2 \times MFD^2$	Corning
K9	Numerical aperture	0.14	NA eff $= 2\lambda/(\pi MFD) = \lambda/(2R_2 \cdot MFD)$	Corning
K10	Cladding diameter	125.0 ± 0.7 μm	Cladding area $= R_2 \times 125^2$	Corning
K11	Attenuation (1550 nm)	≤ 0.18 dB/km	Rayleigh floor $\propto (8R_2/\lambda)^4$	Corning SMF-28 ULL
K12	Cutoff wavelength	1260 nm	$V = 2.405$ at cutoff, $V = 8R_2 \cdot a \cdot NA/\lambda$	ITU-T G.652
K13	Single-mode V-number cutoff	2.405	First zero of J_0	Fiber theory

16.3 Diffraction (Type A)

ID	Formula	Value	R_2/ π Connection
K14	Airy disk first zero	$\sin \theta = 1.22\lambda/D$	$1.22 = 3.8317/ \pi$
K15	First zero of J_1	$j_{11} = 3.83171$	Exact Bessel zero
K16	Airy disk contains	84% of total light in central disk	Exact integral of $[2J_1(x)/x]^2$

ID	Formula	Value	R_2/π Connection
K17	Rayleigh criterion	Two points resolved when separation = $1.22\lambda/D$	$= j_{11}/(\pi) \times \lambda/D$

16.4 Rayleigh Scattering (Type A/M)

ID	Formula	R_2 Form	Precision Data
K18	Scattering cross section	$\sigma \propto (2\pi/\lambda)^4(n^2-1)^2/(n^2+2)^2 \times r^6$	$(8R_2/\lambda)^4$
K19	Fiber Rayleigh coefficient	$\alpha R = A/\lambda^4$	A depends on $(8R_2)^4$
K20	Rayleigh loss floor (1550 nm)	~0.15 dB/km	Set by $(8R_2/1.55)^4$ scattering

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17. Flow, Thermal, and Mechanical (Type A/M)

17.1 R_2 Equations from MATH-1 (the nine engineering domains)

ID	Domain	Formula (Standard)	R_2 Form	Industrial Precision
L1	Pipe flow (area)	$Q = (\pi/4)d^2v$	R_2d^2v	Coriolis: ± 0.05%
L2	Drag force	$F = ½\rho v^2(\pi/4)d^2C\ d$	$½\rho v^2R_2d^2C\ d$	Wind tunnel: ± 1%
L3	Orifice flow	$q = C\ d(\pi/4)d^2\sqrt{(2\Delta P/\rho)}$	$C\ dR_2d^2\sqrt{(2\Delta P/\rho)}$	ISO 5167: ± 0.5%
L4	Capacitor (circular)	$C = \varepsilon_0(\pi/4)d^2/t$	$\varepsilon_0R_2d^2/t$	pF precision
L5	Poynting vector (circular)	$P = S(\pi/4)d^2$	SR_2d^2	Antenna: ± 0.1 dB
L6	Antenna aperture	$A = \eta(\pi/4)D^2$	ηR_2D^2	Calibrated
L7	Beam cross section	$A = (\pi/4)d^2$	R_2d^2	Laser: µm precision
L8	Thermal radiation	$Q = \varepsilon\sigma T^4(\pi/4)d^2$	$\varepsilon\sigma T^4R_2d^2$	Pyrometer: ± 1%

ID	Domain	Formula (Standard)	R_2 Form	Industrial Precision
L9	Sound intensity	$I = P/(4\pi r^2)$	$P/(16R_2r^2)$	SPL meter: $\pm$ 0.5 dB

17.2 Vena Contracta (Type A — exact analytical)

ID	Property	Value	Derivation	Source
L10	Contraction coefficient (2D, sharp-edged)	$C_c = \pi/(\pi+2) = 0.61078\ldots$	Kirchhoff 1869, free-streamline theory	Kirchhoff, J. Math. 70 (1869)
L11	R_2 form	$C_c = 4R_2/(4R_2+2) = 2R_2/(2R_2+1)$	Direct algebraic substitution	—
L12	Discharge coefficient (sharp orifice)	$C_d \approx 0.60$ (= $C_c \times C_v$, $C_v \approx 0.98$)	C_c from theory, C_v from viscous correction	ISO 5167-2:2022
L13	Generalized angle formula	$\Pi_{vc}(\theta) = (\pi/(\pi+2))$^$(2\theta/\pi)$	For any streamline turn angle θ	AguaClara textbook
L14	ISO 5167 database	Reader-Harris/Gallagher equation	C_d from 16,000 calibration points, 9 labs	ISO 5167-2:2022

The vena contracta coefficient C_c = $\pi/(\pi+2)$ is one of the most important R_2-related exact results outside of pure physics. It says: when inviscid fluid exits a sharp-edged orifice, the jet contracts to $\pi/(\pi+2)$ of the orifice area. The number 0.611 that appears in every flow measurement textbook is not an empirical fit — it is an exact analytical result from potential flow theory, derived by Kirchhoff in 1869. In R_2 notation: C_c = $4R_2/(4R_2+2)$.

17.3 Hagen-Poiseuille Flow (Type A)

ID	Formula	Standard	R_2 Form
L15	Volume flow rate	$Q = \pi d^4\Delta P/(128\mu L)$	$R_2d^4\Delta P/(32\mu L) = R_4 \times d^4\Delta P/(\mu L)$

Note: $\pi/128 = (\pi/4)/32 = R_2/32$. And $R_2/32 = \pi/(4 \times 32) = \pi/128$. Also $R_4 = \pi^2/32$, so Hagen-Poiseuille does NOT cleanly factor into R_4. Rather, it is $R_2/32 = R_2 \times (1/32)$.

The Stefan-Boltzmann constant has $R_4 = \pi^2/32$ in it. These are different.

17.4 Other Exact (Type E/A)

ID	Constant	Value	Fraction	Source
L16	Standard gravity g n	9.80665 m/s^2	980665/100000	CGPM 1901
L17	Pendulum formula	$T = 2\pi\sqrt{(L/g)}$	$8R_2\sqrt{(L/g)}$	Mechanics
L18	Polar moment (solid circle)	$J = \pi d^4/32$	$R_2d^4/8$	Structural
L19	Section modulus (solid circle)	$S = \pi d^3/32$	$R_2d^3/8$	Structural

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18. Metrology and Precision Instruments (Type E/M)

ID	Standard/Measurement	Value	Precision	R_2 Connection
M1	Quantum Hall integer plateaus	$R K/i = (8R_2\hbar/e^2)/i$	0.2 ppb	R_2 in R K
M2	Josephson voltage steps	$V n = nhf/(2e) = n/(K J)$	10^{-10}	R_2 in K J via h
M3	Crystal oscillator stability (DOCXO)	$\sim 5 \times 10^{-12}$	$\Delta f/f$	$f = 1/(8R_2\sqrt{(LC)})$
M4	CMM uncertainty (high-end)	~50 nm over 1 m	Linear	—
M5	Surface roughness Ra series	0.1, 0.2, 0.4, 0.8, 1.6, 3.2 µm	Defined nominal	ISO 4287
M6	Gauge block precision	Sub-µm	Interferometric	ISO 3650
M7	Coordinate units	1 inch = 127/5 mm	Exact	Int'l agreement

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19. Geodesy and Navigation (Type E)

ID	Constant	Value	Type	Source
N1	WGS 84 semi-major axis	6378137 m	Exact integer	WGS 84
N2	WGS 84 inverse flattening	298.257223563	Exact rational (9 decimal places)	WGS 84
N3	Standard gravity	9.80665 m/s^2	Exact rational	CGPM 1901
N4	GPS seconds per week	604800	Exact integer	GPS ICD
N5	Earth angular velocity	7.292115×10^{-5} rad/s	Defined	WGS 84

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20. Signal Processing and Mathematical Constants (Type A)

These are exact mathematical relationships where R_2 or R_4 enters the normalization or transform.

ID	Formula	Standard Form	R_2 Form	Usage
O1	Fourier transform norm	$1/(2\pi)$	$1/(8R_2)$	All spectral analysis
O2	Sinc function	$\sin(\pi t)/(\pi t)$	$\sin(4R_2 t)/(4R_2 t)$	Sampling theory
O3	Gaussian normalization	$1/\sqrt{(2\pi)}$	$1/\sqrt{(8R_2)}$	Probability, statistics
O4	Stirling approximation	$n! \approx \sqrt{(2\pi n)}(n/e)^n$	$\sqrt{(8R_2 n)}(n/e)^n$	Combinatorics
O5	Maxwell-Boltzmann speed	$f(v) \propto v^2\exp(-mv^2/2kT)$	Normalization contains $(8R_2)$^(3/2)	Kinetic theory
O6	Euler's identity	e^($i\pi$) = −1	e^($4iR_2$) = −1	Foundational
O7	Basel series	$\Sigma 1/n^2 = \pi^2/6$	$32R_4/6 = 16R_4/3$	Number theory
O8	Wallis product	$\pi/2 = \Pi(4n^2/(4n^2-1))$	$2R_2 = \Pi(4n^2/(4n^2-1))$	Analysis
O9	Gamma(1/2)	$\Gamma(1/2) = \sqrt{\pi}$	$2\sqrt{R_2}$	Special functions

ID	Formula	Standard Form	R_2 Form	Usage
O10	Gauss integral	$\int\exp(-x^2)dx = \sqrt{\pi}$	$2\sqrt{R_2}$	Probability, QFT

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21. BCS Superconducting Gap (Type A/M)

ID	Property	Exact Value	Numerical	Precision Data
O11	BCS gap ratio (weak coupling)	$\Delta_0/(k\ BT\ c) = \pi\ /e\hat{}\gamma$	1.76388…	Exact from BCS theory
O12	Full gap: $2\Delta_0/(k\ BT\ c)$	$2\ \pi\ /e\hat{}\gamma$	3.52757…	—
O13	Euler-Mascheroni constant	γ	0.5772156649…	Computed to 10^9 digits
O14	Al measured gap ratio	—	1.764	Matches BCS to 4 sf
O15	Sn measured gap ratio	—	1.764	Weak-coupling material
O16	Pb measured gap ratio	—	2.185	Strong-coupling deviation (+24%)
O17	Nb measured gap ratio	—	1.87	Intermediate coupling

Significance: this is a non-QED context where π and the Euler-Mascheroni constant γ appear together in an exact, universal, material-independent ratio. γ is in our MATH-2 Tier 2 basis. Computing $\pi\ /e\hat{}\gamma$ in Fraction arithmetic would cross-verify the integer representation of γ against condensed matter measurements.

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22. R_2 Cancellation Identities

These are exact algebraic results where R_2 appears in intermediate steps but cancels in the final measurable quantity. The observation is that the highest-precision measurements in physics are systematically R_2-independent.

ID	Identity	R_2 Status	Precision of Measurement
R1	$K_J \times R_K = 2e/h \times h/e^2 = 2/e$	Cancels	Verified to 10^{-8}
R2	$G_0 \times R_K = 2e^2/h \times h/e^2 = 2$	Cancels	Exact by construction
R3	$R\infty = \alpha^2 m_e c/(2h)$	Cancels ($2h = 16R_2\hbar$)	13 digits
R4	$a_0 = \hbar/(m_e c\alpha) = h/(8R_2 m_e c\alpha)$	Cancels if written as $a_0\alpha = \hbar/(m_e c)$	12 digits
R5	$E_h = m_e c^2\alpha^2$	No R_2	10 digits
R6	Magnetic flux quantum: $\Phi_0^2/R_K = h^2/e^2 \times e^2/h = h$	R_2 reappears	Exact

Structural theorem: define an observable as "R_2-free" if its expression in terms of {h, e, c, m_e, α, …} contains no net factor of π (equivalently, no net factor of R_2) after all cancellations. Theorem: the R_2-free observables are exactly those measurable to 10^{-10} or better. R_2-dependent observables (engineering measurements involving circular cross-sections, solid angles, or loop integrals) are limited to 10^{-12} at best (crystal oscillators, QHE) and typically 10^{-3} to 10^{-6}.

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23. Structural Observations

S1. R_2 universality. $R_2 = \pi/4$ appears in the governing equation of every circular-to-rectilinear conversion. This paper catalogues 20+ independent equations across 17 domains, every one containing R_2 or R_4 in the same structural position: as the conversion factor between d^2 (rectilinear bounding area) and the actual circular area or solid angle.

S2. The vena contracta is R_2-derived. $C\_c = \pi/(\pi+2) = 4R_2/(4R_2+2)$. This is an exact analytical result from 1869 that underpins the ISO 5167 orifice flow standard. Billions of flow meters worldwide implicitly confirm R_2 at their operating precision ($\pm$ 0.5%) every time they compute a flow rate.

S3. The AWG wire gauge system was designed to avoid R_2. By defining the "circular mil" as d^2 rather than $\pi d^2/4$, the AWG system moves R_2 into the conversion factor between circular mils and actual area. The entire system is an engineering workaround to avoid dividing by $4/\pi$ in every wire calculation.

S4. Fiber optics has four independent R_2 appearances. Mode field area ($R_2 \times MFD^2$), effective NA ($\lambda/(2R_2 \times MFD)$), V-number ($8R_2 \times a \times NA/\lambda$), and Rayleigh scattering ($(8R_2/\lambda)^4$). No other single technology concentrates this many independent R_2 appearances.

S5. The Airy constant 1.22 = j_{11}/π . The Rayleigh diffraction limit of every optical system — telescope, microscope, camera, lithography stepper — is set by the ratio of the first Bessel zero to π . This is the circular-aperture analogue of the sinc function's π in the rectangular case.

S6. BCS gap extends the transcendental basis. The superconducting gap ratio π /e^γ involves the Euler-Mascheroni constant γ, which is in our MATH-2 Tier 2 (computable but not yet computed in Fraction arithmetic). This is a non-QED application of the same transcendental basis.

S7. Hagen-Poiseuille carries R_2 into fluid dynamics. The laminar flow equation has prefactor π /128 = R_2/32. This is distinct from $R_4 = \pi^2/32$ in Stefan-Boltzmann. The two R_2 powers (linear in Poiseuille, quadratic in Stefan-Boltzmann) reflect the geometric origin: one cross-section integration in flow, two in radiation.

S8. The R_2 cancellation theorem. The most precisely measured constants (R∞ at 10^{-12}, a_e at 10^{-13}, g-2 at 10^{-10}) are systematically R_2-independent. R_2-dependent measurements (engineering, solid angle, cross-section) are limited to 10^{-12} at absolute best. This suggests the geometric content is "transparent" — it organizes structure but does not limit measurement precision. Level 1 (structure) is confirmed by R_2 universality. Level 2 (parameters) lives in the R_2-free quantities.

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24. Summary Statistics

Category	Entries	Type E	Type S	Type A	Type M
SI Fundamental	7	7	—	—	—
Derived Exact	8	8	—	—	—
Measured Fundamental	12	—	—	—	12
Electroweak	12	—	—	—	12
Quarks/CKM	12	—	—	—	12
Nuclear/Hadron	9	—	—	—	9
Spectroscopy/Clocks	7	—	—	—	7
Optical Storage	31	—	31	—	—
RAM/Memory	13	—	13	—	—
SSD/Storage	11	—	11	—	—
Wire/Cable	23	3	10	2	8
Acoustics/Audio	25	4	7	10	4
RF/Telecom	15	—	10	5	—
Semiconductor	9	—	5	4	—
Optics/Photonics	20	—	—	13	7
Flow/Thermal/Mech	19	1	—	15	3
Metrology	7	1	5	—	1
Geodesy	5	5	—	—	—
Signal Processing	10	—	—	10	—
BCS	7	—	—	3	4

Category	Entries	Type E	Type S	Type A	Type M
R_2 Cancellation	6	—	—	6	—
TOTAL	**~268**	**29**	**92**	**68**	**79**

Of these 268 entries, approximately 90 directly contain R_2 or R_4 in their governing equation. The remainder are inputs to those equations (measured constants), defined standards (sector sizes, transfer rates), or mathematical identities (Fourier, Stirling).

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25. Path to DATA-2

DATA-2 will take every Type E and Type S entry from this inventory and express it as an exact big-integer rational p/q. For Type M entries, DATA-2 will test whether the value at its full measured precision admits a compact rational representation in a Q335 = 2^{335} basis (or another basis to be determined by systematic search). Entries where the Q335 representation is compact (small numerator relative to denominator) are candidates for further investigation. Entries where it is not compact confirm the Level 2 independence established by DISC-6 through DISC-9.

The specific questions for DATA-2:

1. Do all Type E entries have exact Q335 representations? (They should, by construction.)
2. Do Type S entries (engineering nominals) factor cleanly into small primes? (Many will, since they are defined as terminating decimals.)
3. Do any Type M entries (measured constants) have suspiciously compact Q335 representations? (72/72 prior PSLQ tests say no, but the expanded dataset may reveal patterns.)
4. Is Q335 = 2^{335} the optimal basis, or does a different power of 2 (or a different prime base) give better factorizations across the full dataset?

HOWL-DATA-1 is a living document. Entries will be added as new domains are searched and new precision data becomes available. The current version catalogues 268 entries across 17 domain categories, with 90+ direct R_2/R_4 connections.

[**HOWL-DATA-1-2026**] Cross-Domain Precision Data Inventory. Github: https://github.com/ghowland/papers/tree/main/DATA-1-2026

[**HOWL-DATA-2-2026**] Integer Rational Representations in Q335. Github: https://github.com/ghowland/papers/tree/main/papers/DATA/HOWL-DATA-2-2026